

Mathematics - Probability and Statistics

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Class details

- 35h: 2h*15+ 2h mid term +3h final exam
- Grade:
 - 35% Mid term (2h)
 - 65% Final Exam (3h)
- For anonymous feedback: [Feedback - Google Docs](#)

Class organisation

- Lectures (~1h)
- Exercices (~30min)
- Labs (~30min)

Syllabus

- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
- **Session 7:** Total Probability & Bayes' Theorem
- **Session 8:** Discrete Random Variables
- **Session 9:** Continuous Random Variables
- **Session 10:** Joint Distributions & Correlation
- **Session 11: Exercices/Summary**
- **Session 12:** Central Limit Theorem & Simulation
- **Session 13:** Point Estimation & MLE
- **Session 14:** Confidence Intervals
- **Session 15:** Hypothesis Testing – Basics

Environment & Requirements

Tools & Libraries to Install

Library

numpy

matplotlib

Pandas

You can run this course on **Google Colab** or **locally**.

Google Colab

What is Google Colab? Google Colab is a free cloud service that supports Jupyter notebooks and provides free access to computing resources including GPUs.

To use it:

- Log in with a Google account
- Open a .ipynb notebook from Drive or GitHub, or create a new one
- To enable GPU: Runtime > Change runtime type > GPU

Why we use it:

- Free and powerful
- No local install needed
- Familiar interface for Jupyter users

Upload files as a drive from your local machine

```
from google.colab import files  
input_file = files.upload()
```

Bibliography

- <https://online.stat.psu.edu/stat414/>
- [Probability & Statistics for Machine Learning and Data Science](#)
- [Applied Statistics and Probability for Engineers, 5e](#)
- [statbook3.pdf](#)
- [Statistics and Probability Full Course || Statistics For Data Science - YouTube](#)

Why Study Statistics?

Session 1

Why Study Statistics?

Session 1

Objectives:

- Understand the goals of statistics,
- Differentiate between populations and samples,
- Distinguish between parameters and statistics,
- Classify types of variables and levels of measurement,
- Recognize the challenges of sampling and why randomness is essential,
- And understand why we need to quantify uncertainty

The Origins of Statistics

- **Statistics" comes from Latin status, and Italian Renaissance usage relating to the “state”**
- First systematic collections of population and economic data occurred in Venice and Florence during the Renaissance
- By the 16th century, many European governments required parishes to record:
 - Births
 - Marriages
 - Deaths



The first “big data” analysis in England was used to help the king avoid the **plague** — basically the original version of **Google Maps alerts**.

So What Exactly Is Statistics?

Statistics is the branch of mathematics concerned with:

- **Collecting** data
- **Organizing & summarizing** data
- **Analyzing** data to find patterns or relationships
- **Interpreting** results to draw conclusions
- **Communicating** findings clearly

The goal: Turn **data** into **insight**, and **uncertainty** into **informed decisions**

Statistics Is Everywhere

Statistics is used in **every field that collects and interprets data**:

- **Science** – medical studies, climate models, physics experiments
- **Business** – customer analytics, market research, A/B testing
- **Technology** – AI, machine learning, recommendation systems
- **Politics** – opinion polling, policy analysis, voting behavior
- **Health** – epidemiology, vaccine effectiveness, risk modeling
- **Social sciences** – psychology, economics, education
- **Sports** – performance stats, player evaluation, predictions

Why We Need Statistics: Our Brain Can Be Misleading

Our intuition is biased:

Availability bias – we judge frequency by what's easy to recall
“Shark attacks are more memorable than car accidents”

Confirmation bias – we seek facts that fit our beliefs

Overconfidence bias – we think we know more than we do

Statistics helps us:

- See **through randomness**
- Make decisions based on **evidence**, not impressions
- Communicate **uncertainty** clearly

How Does a Statistician Think?

The 5 Steps of the Statistical Process:

- **Ask a clear question**
 - What do we want to know?
- **Collect data**
 - Design an experiment or survey
- **Describe the data**
 - Use graphs and numerical summaries
- **Analyze the data**
 - Use probability to model and test
- **Interpret and communicate results**
 - Draw conclusions and report uncertainty

Population vs Sample – Who Are We Studying?

Definition

In statistics, a population is the set on which a statistical study is conducted. A sample is a (usually known) subset of the population

Population

→ The **entire group** we are interested in studying

Examples: all French voters, all EPITA students, all galaxies in the observable universe

Sample

→ A **subset** of the population that we actually collect data from

Examples: 1,000 survey respondents, 50 students, 10,000 galaxies from SDSS

Why use a sample?

- **Faster**
- **Cheaper**
- Often **more practical or ethical**

Goal: Make **valid conclusions** about the population **from** the sample

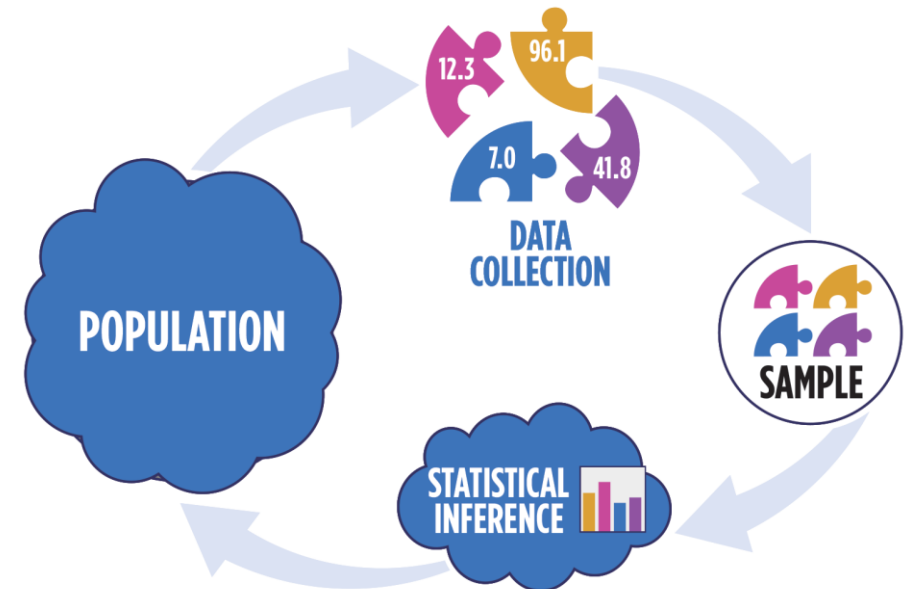
Descriptive vs Inferential Statistics

Descriptive Statistics

- Summarize and describe observed data
- Concerned with the **sample** only
- Probabilities play only a minor role here.
- Tools: mean, median, standard deviation, histograms, etc.

Inferential Statistics

- Use data from an observed sample to make a conclusion about a population based on a sample
- Tools: estimation, hypothesis tests, confidence intervals



Eugene Morgan © Penn State

Population vs Sample – Who Are We Studying?

**Is this classroom a good sample
of all EPITA students?**

Parameters vs Statistics

Definition:

A parameter is a fixed number that describes the population — like the true average age of all EPITA students.

But we usually can't measure it, so we estimate it using a statistic — like the average age in a sample of 100 students."

Population mean = μ

Sample mean = $\bar{x}$

Population standard deviation = σ

Sample standard deviation = s

Statistic is a random variable — if you take another sample, you'll get a different number. That's why we later use confidence intervals to quantify the uncertainty.

What is a variable?

A variable is a characteristic, property, or quantity that can be measured or observed and that varies from one individual or unit to another.

In other words:

- A variable can take different values across a population
- It can be something you ask about, observe, or measure

Examples of Variables:

- Student age
- Favorite programming language
- GPA
- Height
- City of birth

Types of Variables – What Kind of Data Are We Dealing With?

Two Main Types of Variables:

1. Categorical (Qualitative)

- Describes qualities or categories
- Cannot be meaningfully measured or averaged

Examples:

- Gender
- Eye color

Subtypes:

- Nominal: no order (e.g., eye color)
- Ordinal: with order (e.g., satisfaction level)

2. Numerical (Quantitative)

- Represents measurable quantities
- Can be used in mathematical operations

Examples:

- Age
- Height

Subtypes:

- Discrete: countable (e.g., exams passed)
- Continuous: measurable (e.g., weight, time)

Types of Variables – What Kind of Data Are We Dealing With?

Examples:

For the following examples, do you think they are discrete qualitative or quantitative or continuous quantitative variables?

- voting in an election;
- weight;
- rolling a die;
- amount of chocolate consumed in a year;
- being left-handed or right-handed;

Types of Variables – What Kind of Data Are We Dealing With?

Examples:

For the following examples, do you think they are discrete qualitative or quantitative or continuous quantitative variables?

- voting in an election; **Qualitative – Nominal** (e.g., Yes/No or candidate names — no inherent order)
- weight; **Quantitative – Continuous** (Can take any value within a range, with decimals)
- rolling a die; **Quantitative – Discrete** (Only whole number outcomes: 1–6)
- amount of chocolate consumed in a year; **Quantitative – Continuous** (Can be measured in grams or kilograms — decimals possible)
- being left-handed or right-handed; **Qualitative – Nominal** (Categories with no natural order)

The Four Levels of Measurement

Not all data are created equal. Some variables just tell us names or categories, while others allow us to compute things like averages and ratios. To choose the right statistical methods, you must first understand the **level of measurement of your variable**. In contrast to nominal and ordinal scale variables, **interval scale and ratio scale** variables are variables for which the numerical value is genuinely meaningful

- Interval: differences between the numbers are interpretable, but the variable doesn't have a "natural" zero value.

Level	Ordered?	Equal Intervals?	True Zero?	Example
Nominal	✗ No	✗ No	✗ No	Eye color, blood type
Ordinal	✓ Yes	✗ No	✗ No	Satisfaction level, ranking
Interval	✓ Yes	✓ Yes	✗ No	Temperature in °C, IQ score
Ratio	✓ Yes	✓ Yes	✓ Yes	Height, age, income

The Four Levels of Measurement

Variable	Type?	Level of Measurement?
Blood Type		
Movie Rating (★ to ★★★★★)		
Temperature in °C		
Distance Run (in km)		
T-shirt Size (S, M, L, XL)		
Postal Code		
Age		
Exam Score (%)		
Time of Day (24h clock)		
Pain Level (1 to 10)		

The Four Levels of Measurement

Variable	Type?	Level of Measurement?
Blood Type	Categorical	Nominal
Movie Rating (★ to ★★★★★)	Categorical	Ordinal
Temperature in °C	Numerical	Interval
Distance Run (in km)	Numerical	Ratio
T-shirt Size (S, M, L, XL)	Categorical	Ordinal
Postal Code	Categorical	Nominal
Age	Numerical	Ratio
Exam Score (%)	Numerical	Ratio
Time of Day (24h clock)	Numerical	Interval
Pain Level (1 to 10)	Categorical*	Ordinal

The Dangers of Sampling

Having data doesn't mean it's good data.

A common mistake is to believe that any large dataset is automatically valid — not true. If the sample is biased, your results will be flawed.

Exemple:

One student wants to prepare a cake for this class (chocolate or lemon) of 40 students. He asks to 10 friends:

- Is it representative?

Now she asks for a votation on a political subject. She asks again 10 friends

-Is it representative?

Real-world cautionary tales:

- Literary Digest poll in 1936 (2.2 millions, wrong result) vs George Gallup using 50 000 persons
- Brexit
- 2016 Trump election

Solutions

Common Pitfalls:

- **Biased samples**
 - Only ask your friends? Expect biased answers
- **Self-selection**
 - Only people with strong opinions tend to respond
- **Too small / not diverse**
 - A tiny group may not reflect the larger population

In an ideal world, we want every individual in the population to have an equal chance of being selected. That's what we call random sampling — and it's the gold standard

Unfortunately, random sampling is not always easy..."

- You need a **complete list** of the population
- You need **consent** from participants
- You need **time and resources**

Quantify the uncertainty

Finally, **statistics is not about giving one answer**. It's about saying:

Here's what we think — and how confident we are.'

We need to provide:

- A margin of error
- A confidence interval
- An estimate of variance

A result without a measure of uncertainty is scientifically meaningless.

Exercices

What's the population if I want to study social media usage in France?

What would be a representative sample?

You survey only your Instagram followers. What type of bias might this introduce?

What type of variable is *number of hours spent online per day*?

Exercices

Population (target): All people living in France (the residents you want to draw conclusions about).

Representative sample: A probability sample of residents of France stratified by age, gender, region, urban/rural, and socioeconomic status, then randomly selected.

Surveying only your Instagram followers bias: It overrepresents Instagram users who follow you (similar demographics/interests) and undercovers non-Instagram users and those outside your network.

Variable type – “number of hours online per day”: Quantitative, continuous (ratio scale).

Exercices

An election will be held next week and, by polling a sample of the voting population, we are trying to predict whether the Republican or Democratic candidate will prevail. Which of the following methods of selection is likely to yield a representative sample?

- (a) Poll all people of voting age attending a college basketball game.
- (b) Poll all people of voting age leaving a fancy midtown restaurant.
- (c) Obtain a copy of the voter registration list, randomly choose 100 names, and question them.
- (d) Use the results of a television call-in poll, in which the station asked its listeners to call in and name their choice.
- (e) Choose names from the telephone directory and call these people.

Exercices

An election will be held next week and, by polling a sample of the voting population, we are trying to predict whether the Republican or Democratic candidate will prevail. Which of the following methods of selection is likely to yield a representative sample?

(a) Poll all people of voting age attending a college basketball game.

Biased. You're likely to get younger, more sport-oriented individuals, not a broad cross-section of the population.

(b) Poll all people of voting age leaving a fancy midtown restaurant.

Biased. This group likely has higher income and may not represent lower or middle-income voters.

(c) Obtain a copy of the voter registration list, randomly choose 100 names, and question them.

Best choice. Random sampling from a list of registered voters ensures every voter has an equal chance of selection — key for representativeness.

(d) Use the results of a television call-in poll, in which the station asked its listeners to call in and name their choice.

Very biased. Voluntary response bias — people who call in are self-selected and more opinionated.

(e) Choose names from the telephone directory and call these people.

Biased. Not everyone is listed in the phone book. Undercoverage bias is likely.

Exercices

The approach used in Problem 1(e) led to a disastrous prediction in the 1936 presidential election, in which Franklin Roosevelt defeated Alfred Landon by a landslide. A Landon victory had been predicted by the Literary Digest. The magazine based its prediction on the preferences of a sample of voters chosen from lists of automobile and telephone owners.

Why do you think the Literary Digest's prediction was so far off ?

Exercices

The approach used in Problem 1(e) led to a disastrous prediction in the 1936 presidential election, in which Franklin Roosevelt defeated Alfred Landon by a landslide. A Landon victory had been predicted by the Literary Digest. The magazine based its prediction on the preferences of a sample of voters chosen from lists of automobile and telephone owners.

Why do you think the Literary Digest's prediction was so far off ?

The Literary Digest's prediction failed because of selection bias — they drew their sample from lists of automobile and telephone owners, **which in 1936 disproportionately represented wealthier Americans, who were more likely to vote Republican.**

In 1936, during the Great Depression, many people couldn't afford phones or cars.

Therefore, the sample underrepresented lower-income groups, who largely supported Roosevelt (Democrat). **Even though the sample size was very large (millions), it was not representative of the general voting population.**